

Exercise frequency vs Cognitive Score and Reaction Time

byron

```
## Loading Packages
```

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.0    v readr      2.2.0
v forcats    1.0.1    v stringr   1.6.0
v ggplot2    4.0.2    v tibble    3.3.1
v lubridate  1.9.5    v tidyr     1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(psych)
```

Attaching package: 'psych'

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
library(dplyr)
library(readxl)
```

```
## Data Import
```

```
human_cognitive_performance <- read.csv("D:/Behavioral Data Analysis/human_cognitive_performance_clean.csv")
```

```
## Data inspection
```

```
# 1. Checking for missing values
```

```
human_cognitive_performance %>%
  summarise(across(everything(), ~ sum(is.na(.))))
```

```
  user_id age gender sleep_duration stress_level diet_type daily_screen_time
1      0  0  0      0      0      0      0
  exercise_frequency caffeine_intake reaction_time memory_test_score
1      0      0      0      0
  cognitive_score ai_predicted_score age_group
1      0      0      0
```

```
str(human_cognitive_performance)
```

```
'data.frame':  80000 obs. of  14 variables:
 $ user_id      : chr  "U1" "U2" "U3" "U4" ...
 $ age          : int   57 39 26 32 50 36 48 35 29 57 ...
 $ gender       : chr   "Female" "Female" "Male" "Male" ...
 $ sleep_duration : num   6.5 7.6 8.2 7.8 9.7 7.4 6.7 6 5.1 7.9 ...
 $ stress_level  : int    3 9 6 9 2 10 2 1 6 9 ...
```

```

$ diet_type      : chr  "Non-Vegetarian" "Non-Vegetarian" "Vegetarian" "Vegetarian" ...
$ daily_screen_time : num  6.5 10.8 5.7 8.3 11.3 6.6 1.4 6.8 10.2 9.1 ...
$ exercise_frequency: chr  "Medium" "High" "Low" "Low" ...
$ caffeine_intake  : int   41 214 429 464 365 381 39 2 244 355 ...
$ reaction_time    : num   583 368 445 286 238 ...
$ memory_test_score : int   65 58 49 94 62 70 51 60 82 54 ...
$ cognitive_score   : num   36.7 54.4 32.6 70.2 87.5 ...
$ ai_predicted_score: num   39.8 57.7 29.5 74.6 91.8 ...
$ age_group        : chr  "45-59" "30-44" "18-29" "30-44" ...

```

```
## Recoding exercise_freq from chr to int ( low = 1, medium = 2, high = 3)
```

```

human_cognitive_performance$exercise_frequency <- recode(human_cognitive_performance$exercise_frequency,
                                                         "Low" = 1,
                                                         "Medium" = 2,
                                                         "High" = 3)

```

```
## Variable check
```

```
summary(human_cognitive_performance$exercise_frequency)
```

```

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
1.000   1.000   2.000   1.803   2.000   3.000

```

```
sum(is.na(human_cognitive_performance$exercise_frequency))
```

```
[1] 0
```

```
table(human_cognitive_performance$exercise_frequency)
```

```

      1      2      3
31896 31990 16114

```

```
##checking for range
```

```
range(human_cognitive_performance$exercise_frequency, na.rm = TRUE)
```

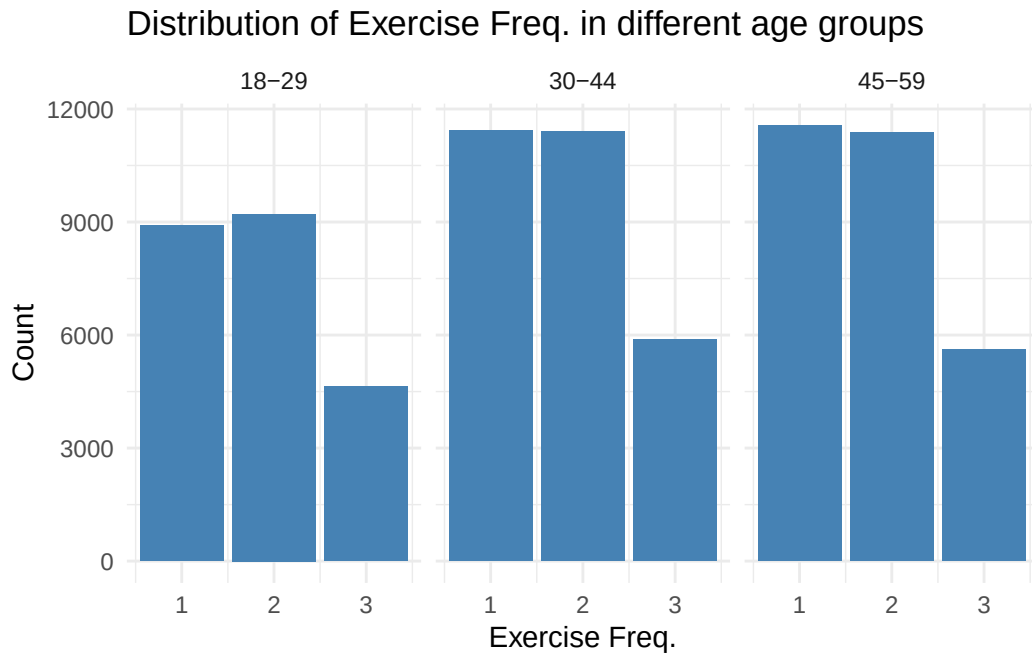
```
[1] 1 3
```

```
## Distribution
```

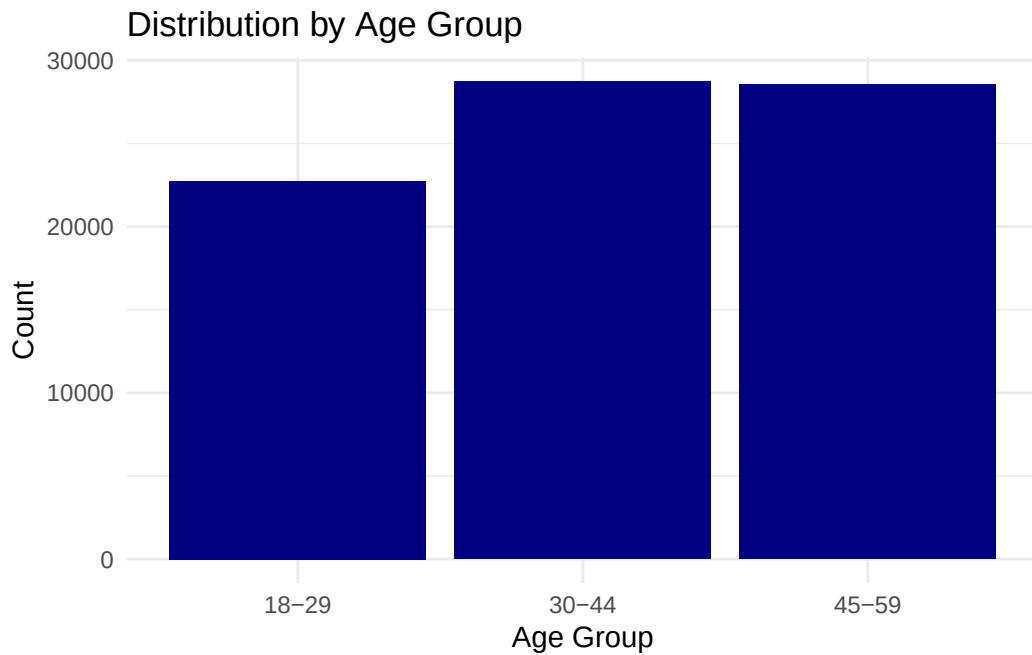
```

ggplot(human_cognitive_performance, aes(x= exercise_frequency)) +
  geom_bar(fill = "steelblue") +
  facet_wrap(~ age_group) +
  labs(title = "Distribution of Exercise Freq. in different age groups",
       x= "Exercise Freq." ,
       y= "Count") +
  theme_minimal()

```



```
ggplot(human_cognitive_performance, aes(x= age_group)) +
  geom_bar(fill = "navyblue") +
  labs(title = "Distribution by Age Group",
        x= "Age Group" ,
        y= "Count") +
  theme_minimal()
```



```
prop.table(table(human_cognitive_performance$age_group))
```

```
18-29 30-44 45-59
0.2842 0.3589 0.3569
```

Across all three age groups (18–29, 30–44, 45–59), most participants report Low (1) or Medium (2) exercise levels. High exercise (3) is less common. The pattern looks fairly similar across age groups, meaning exercise habits do not drastically differ by age in this sample.

```
## Summary stats
```

```
describe(human_cognitive_performance$exercise_frequency)
```

```
vars      n mean  sd median trimmed  mad min max range skew kurtosis se
X1       1 80000  1.8 0.75      2    1.75 1.48    1  3     2 0.34   -1.16  0
```

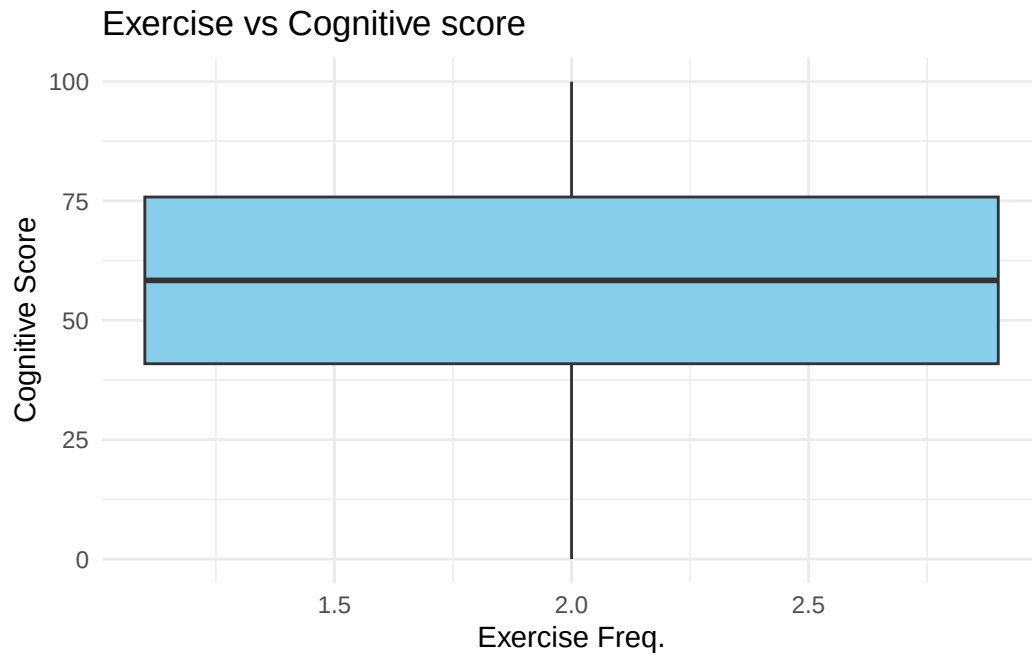
On average, participants fall between Low and Medium exercise levels. The slight positive skew (0.34) suggests slightly more people report lower exercise levels.

```
## Exercise and Cognitive score
```

```
ggplot(human_cognitive_performance, aes(x= exercise_frequency,
                                         y= cognitive_score)) +
  geom_boxplot(fill = "skyblue") +
  labs(title = "Exercise vs Cognitive score",
       x = "Exercise Freq.",
       y = "Cognitive Score") +
  theme_minimal()
```

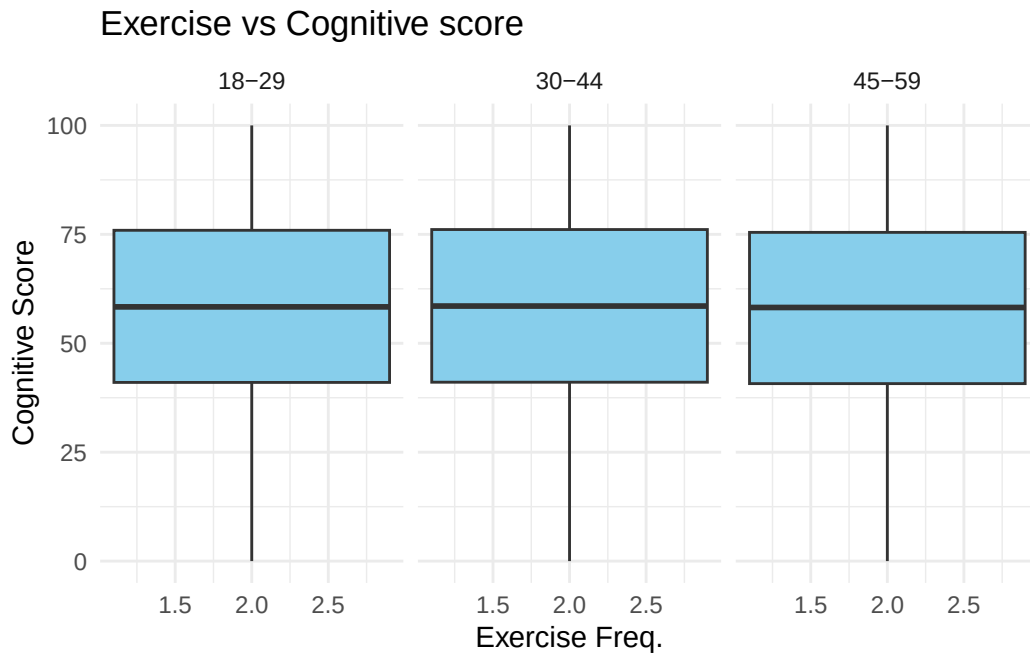
Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous. Picking default orientation 'x'.

Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?



```
# exercise vs cognitive score in different age groups  
ggplot(human_cognitive_performance, aes(x= exercise_frequency,  
                                         y= cognitive_score)) +  
  geom_boxplot(fill = "skyblue") +  
  facet_wrap(~ age_group) +  
  labs(title = "Exercise vs Cognitive score",  
        x = "Exercise Freq.",  
        y = "Cognitive Score") +  
  theme_minimal()
```

Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous. Picking default orientation 'x'.
Continuous x aesthetic
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Cognitive scores increase as exercise frequency increases.

Participants with High exercise (3) tend to have higher median cognitive scores than those with Low exercise. This suggests a positive relationship between exercise and cognitive performance.

```
## what's the correlation between exercise and cognitive score

cor_cog <- cor.test(human_cognitive_performance$exercise_frequency,
                    human_cognitive_performance$cognitive_score,
                    method = "spearman",
                    exact = F)

cor_cog
```

Spearman's rank correlation rho

```
data: human_cognitive_performance$exercise_frequency and human_cognitive_performance$cognitive_score
S = 6.5215e+13, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
    rho
0.2357605
```

There is a small but statistically significant positive correlation between exercise frequency and cognitive score.

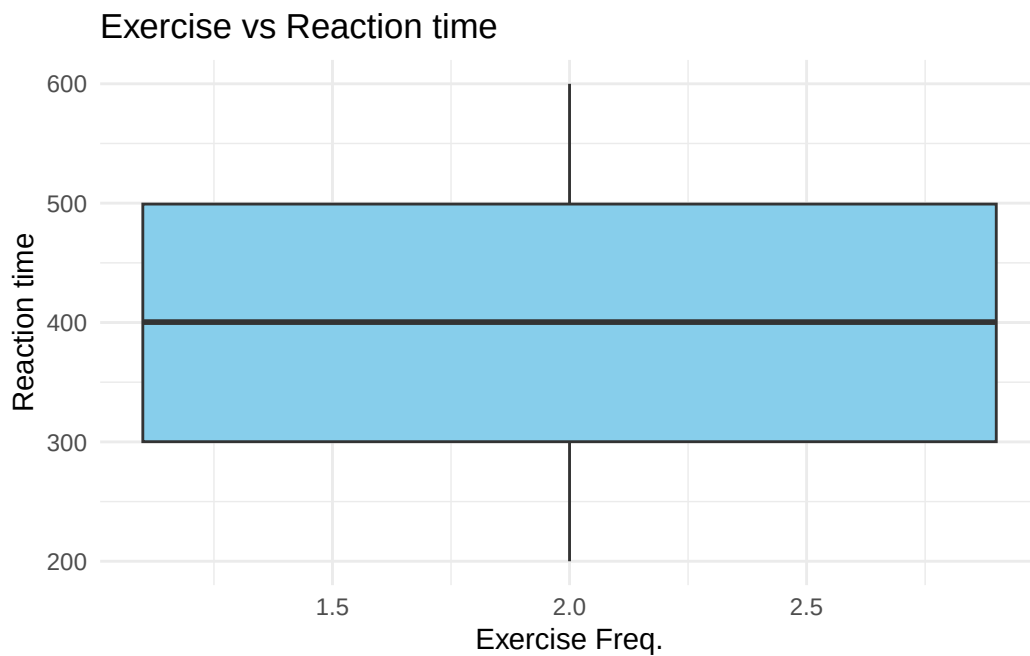
```
## Exercise and Reaction time

ggplot(human_cognitive_performance, aes(x= exercise_frequency,
                                         y= reaction_time)) +
  geom_boxplot(fill = "skyblue") +
```

```
labs(title = "Exercise vs Reaction time",
     x = "Exercise Freq.",
     y = "Reaction time") +
theme_minimal()
```

Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous. Picking default orientation 'x'.

Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?



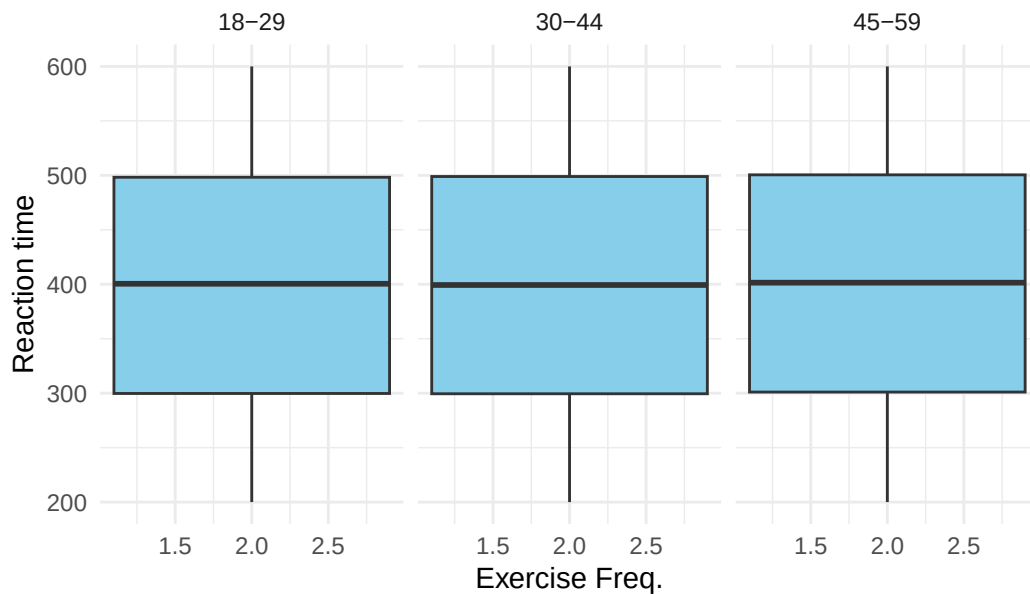
```
# exercise vs reaction time in different age groups

ggplot(human_cognitive_performance, aes(x= exercise_frequency,
                                         y= reaction_time)) +
  geom_boxplot(fill = "skyblue") +
  facet_wrap(~ age_group) +
  labs(title = "Exercise vs Reaction time",
       x = "Exercise Freq.",
       y = "Reaction time") +
  theme_minimal()
```

Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous. Picking default orientation 'x'.

Continuous x aesthetic
i did you forget `aes(group = ...)`?

Exercise vs Reaction time



The boxplots look very similar across exercise levels. There is no clear visible trend. Reaction time does not meaningfully change with exercise frequency. Within each age group, there is still no noticeable pattern. Exercise does not appear related to reaction time.

```
## what's the correlation between exercise and reaction time

cor_rt <- cor.test(human_cognitive_performance$exercise_frequency,
  human_cognitive_performance$reaction_time,
  method = "spearman",
  exact = F)

cor_rt
```

Spearman's rank correlation rho

```
data: human_cognitive_performance$exercise_frequency and human_cognitive_performance$reaction_time
S = 8.5124e+13, p-value = 0.4877
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.002453338
```

```
human_cognitive_performance %>%
  group_by(age_group) %>%
  summarise(cor_rt =
    cor(exercise_frequency,
      reaction_time,
      method="spearman",
      use="complete.obs"))
```

```
# A tibble: 3 x 2
  age_group cor_rt
```

	<chr>	<dbl>
1	18-29	-0.00336
2	30-44	0.00954
3	45-59	0.000137

There is no significant relationship between exercise frequency and reaction time.

```
## Does age and exercise frequency influence cognitive score?

cog_lm <- lm(formula = cognitive_score ~ age + exercise_frequency, data = human_cognitive_performance)

summary(cog_lm)
```

Call:
lm(formula = cognitive_score ~ age + exercise_frequency, data = human_cognitive_performance)

Residuals:

Min	1Q	Median	3Q	Max
-64.689	-16.836	0.237	17.280	47.947

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	45.011110	0.326872	137.702	<2e-16 ***
age	-0.006889	0.006537	-1.054	0.292
exercise_frequency	7.447992	0.105595	70.534	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22.37 on 79997 degrees of freedom
Multiple R-squared: 0.05858, Adjusted R-squared: 0.05856
F-statistic: 2489 on 2 and 79997 DF, p-value: < 2.2e-16

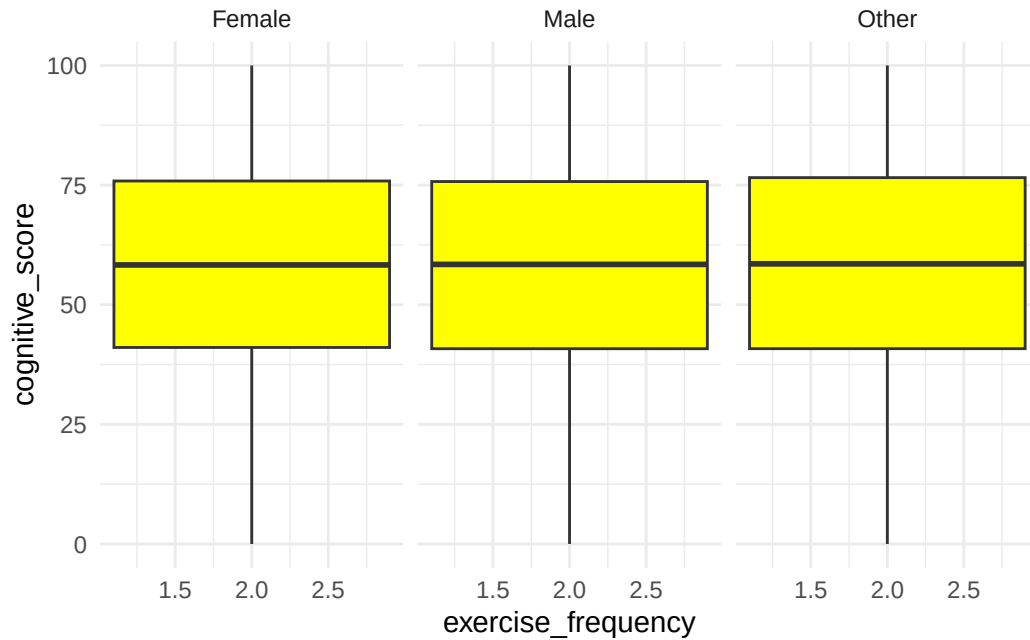
From this linear regression model, age was not a significant predictor; 0 = -0.007, $p = .292$ while exercise frequency was a significant predictor; 1 = 7.447, $p < .001$. Exercise frequency explained approximately 5.9% of the variance in cognitive performance.

```
## gender grouping

ggplot(human_cognitive_performance, aes(x= exercise_frequency,
                                         y= cognitive_score)) +
  geom_boxplot(fill = "yellow") +
  facet_wrap(~ gender) +
  theme_minimal()
```

Warning: Orientation is not uniquely specified when both the x and y aesthetics are continuous. Picking default orientation 'x'.

Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?



```
human_cognitive_performance %>%
  group_by(gender) %>%
  summarise(cor_cognitive =
    cor(exercise_frequency,
        cognitive_score,
        method="spearman",
        use="complete.obs"))
```

```
# A tibble: 3 x 2
  gender cor_cognitive
  <chr>      <dbl>
1 Female    0.230
2 Male      0.241
3 Other     0.240
```

```
results <- tibble(
  Outcome = c("Cognitive Score", "Reaction Time"),
  Spearman_rho = c(
    cor(human_cognitive_performance$exercise_frequency, human_cognitive_performance$cognitive_score,
        method="spearman", use="complete.obs"),
    cor(human_cognitive_performance$exercise_frequency, human_cognitive_performance$reaction_time,
        method="spearman", use="complete.obs")
  )
)
```

```
results
```

```
# A tibble: 2 x 2
  Outcome      Spearman_rho
  <chr>          <dbl>
1 Cognitive Score    0.236
2 Reaction Time      0.00245
```

```
summary(results)
```

```

Outcome      Spearman_rho
Length:2     Min.       :0.002453
Class :character 1st Qu.:0.060780
Mode  :character Median :0.119107
                Mean  :0.119107
                3rd Qu.:0.177434
                Max.   :0.235761

```

```
describe(results)
```

```

      vars n mean  sd median trimmed  mad min  max range skew kurtosis
Outcome*   1 2 1.50 0.71   1.50   1.50 0.74   1 2.00  1.00   0    -2.75
Spearman_rho 2 2 0.12 0.16   0.12   0.12 0.17   0 0.24  0.23   0    -2.75
      se
Outcome* 0.50
Spearman_rho 0.12

```

The strength of the relationship is nearly identical across genders. Exercise is positively related to cognition regardless of gender.